
Attribution Method Fingerprinting: Characterizing Data Attribution via Contrastive Mode Probing

Anonymous Authors¹

Abstract

Data attribution methods identify which training examples influenced a model’s predictions, yet leading methods—TRAK, TracIn, and Kronfluence—often disagree substantially on the same examples. This work investigates whether such disagreement reflects systematic differences in what each method measures. Using contrastive mode probing, mode profiles capturing sensitivity to memorization, feature transfer, and spurious association are computed for each method. Experiments on CIFAR-10 with ResNet-18, ViT-Small, and ConvNeXt-Tiny reveal that methods are strongly dissociable ($F=1423.55$, $p=4.3 \times 10^{-28}$) with large effect sizes (Cohen’s $d=20.64$), and that mode profiles exhibit high internal consistency (Cronbach’s $\alpha>0.96$ for all methods). However, cross-architecture transfer is limited: profiles correlate positively within architectural families (ResNet-ViT $r=0.80$) but invert across fundamentally different architectures (ConvNeXt shows $r=-0.71$ to -0.99 versus CNN/Transformer). These findings indicate that attribution method fingerprints are reliable within architectural families but require calibration for cross-family comparisons. The results suggest reframing attribution method selection from “which method wins” to “what does each method measure.”

1. Introduction

When three data attribution methods—TRAK, TracIn, and Kronfluence—are applied to identify which training examples influenced a model’s predic-

tion, they can produce substantially different rankings. This raises a question: are these methods measuring the same phenomenon, or do they capture fundamentally different aspects of training data influence?

This question has practical implications. Data attribution underpins applications including identifying training examples that cause model failures, auditing models for fairness violations, and valuing data contributions. If different methods systematically emphasize different aspects of influence—memorization versus feature transfer versus spurious correlation—then method choice shapes the conclusions practitioners draw.

The DATE-LM benchmark (Jiao et al., 2025) confirmed that no single attribution method dominates across all tasks. TRAK performs well on some evaluations while influence functions perform better on others. However, this finding describes that methods differ without explaining why they differ or what each method measures.

This work investigates whether unexplained disagreement reflects structure: different attribution algorithms compute influence via mathematically distinct operations that create characteristic “fingerprints.” TRAK uses random projection of gradients, emphasizing direction alignment in parameter space. TracIn sums checkpoint-weighted gradient dot products, capturing temporal patterns across training. Kronfluence inverts Fisher information via K-FAC approximation, emphasizing curvature in the loss landscape. These operations are not minor variations—they encode different assumptions about what makes training data influential.

A framework is developed for characterizing attribution methods by their mode profiles: systematic patterns of sensitivity to memorization, feature transfer, and spurious association. If methods embed different biases, mode profiles should dissociate—methods should cluster by identity rather than by random variation. If fingerprints are intrinsic to algorithms rather than artifacts of specific datasets, they should exhibit

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

stability within methods.

The contributions are:

First, contrastive mode probing is introduced as a methodology for measuring attribution method sensitivity to specific influence modes using test cases that isolate memorization, feature transfer, and spurious association signals.

Second, quantitative characterization of method dissociation is provided. Using ANOVA analysis across 10 independent runs per method, inter-method variance exceeds intra-method variance with $F=1423.55$ (threshold: 4.0), and effect sizes are large (Cohen’s $d=20.64$).

Third, transferability of mode profiles across architectures is analyzed. Fingerprints transfer within architectural families (ResNet-ViT $r=0.80$) but invert across fundamentally different architectures (ConvNeXt shows $r=-0.71$ to -0.99 versus CNN/Transformer). This finding bounds the generalization of fingerprinting.

2. Related Work

2.0.1. DATA ATTRIBUTION METHODS

Training data attribution has evolved from influence functions (Koh & Liang, 2017) through increasingly scalable approximations. TracIn (Pruthi et al., 2020) introduced gradient tracing across checkpoints. TRAK (Park et al., 2023) achieved speedup through random projection of gradients. Kronfluence (Grosse et al., 2023) applied Kronecker-factored curvature approximation (K-FAC) to enable influence computation at scale.

These methods represent different computational strategies, but prior work has not systematically characterized what each strategy measures. This work addresses this gap by showing that computational differences manifest as systematic differences in sensitivity to influence modes.

2.0.2. ATTRIBUTION METHOD FRAGILITY AND BENCHMARKING

Basu et al. (2020) demonstrated that influence functions are fragile in deep networks, with approximation errors increasing with network depth. This finding raised concerns about gradient-based attribution reliability. The present work suggests that apparent fragility may be reframed: rather than random errors, different methods may exhibit systematic sensitivity patterns that constitute predictable fingerprints.

The DATE-LM benchmark (Jiao et al., 2025) provided the first unified evaluation of LLM attribution methods, revealing that no single method dominates across tasks. This work complements DATE-LM by investigating why methods differ: the inductive biases embedded in each algorithm’s mathematics may create different sensitivities to memorization, feature transfer, and spurious association.

Several efficiency-focused works have pushed attribution to larger scales. LoRIF (Li et al., 2026) achieved $20\times$ storage reduction on 70B models, and GraSS (Hu et al., 2025) demonstrated 165% throughput improvement through gradient sparsification. GGDA (Ley et al., 2024) introduced group-level attribution with $10\text{-}50\times$ speedup. These works focus on computational efficiency; this work focuses on characterizing what methods measure.

2.0.3. LEARNING DYNAMICS AND INFLUENCE MODES

Research on learning dynamics has identified distinct phases and patterns in neural network training. Studies of memorization (Feldman, 2020) show that models memorize atypical examples to achieve low training loss. Work on spurious correlations (Sagawa et al., 2020) reveals that models exploit dataset shortcuts.

These distinct influence modes—memorization, feature transfer, spurious association—have been studied in isolation. The contrastive mode probing framework provides a unified lens for measuring how attribution methods respond to each mode.

3. Methodology

3.0.1. OVERVIEW

Given an attribution method A , a trained model M , and a set of contrastive probe pairs P , the method’s sensitivity to each influence mode is measured by computing attribution scores on probes designed to isolate that mode. The resulting mode profile is a vector $p_A = [s_{\text{mem}}, s_{\text{transfer}}, s_{\text{spurious}}]$ capturing the method’s relative sensitivity to memorization, feature transfer, and spurious association.

3.0.2. INFLUENCE MODES

Three modes of training data influence are defined:

Memorization: Training examples influence test predictions through direct memorization—the model learns the specific input-output mapping rather than generalizable features. Attribution methods sensitive to memorization assign high influence to near-

duplicate or highly similar training examples.

Feature Transfer: Training examples contribute features that generalize to test examples in the same category. Attribution methods sensitive to feature transfer assign influence based on shared high-level representations rather than surface similarity.

Spurious Association: Training examples share spurious correlations (e.g., background features, co-occurring attributes) with test examples. Attribution methods sensitive to spurious association may attribute influence to examples sharing these shortcuts.

3.0.3. CONTRASTIVE PROBE CONSTRUCTION

For each mode $m \in \text{mem}$, transfer, spurious, contrastive probe pairs (t, s^+, s^-) are constructed where t is a test point, s^+ is a training point exhibiting mode m relationship with t , and s^- is a matched control lacking this relationship.

Memorization probes: s^+ selected as training examples with high pixel-level similarity to t (near-duplicates). s^- selected from the same class with low surface similarity.

Feature transfer probes: s^+ selected from the same semantic class as t but with different surface features. s^- selected from a different class with similar surface statistics.

Spurious probes: s^+ selected sharing a spurious correlation with t . s^- selected from the same class without the spurious correlation.

Mode sensitivity is computed as:

$$s_m = (1/|P_m|)\Sigma \text{sign}(A(s^+; t) - A(s^-; t))$$

where $A(s; t)$ is the attribution score for training point s given test point t .

3.0.4. DISSOCIATION ANALYSIS

To test whether methods produce dissociable fingerprints, each method is run with multiple random seeds and ANOVA analysis is applied:

Between-group variance (inter-method): Variance of method means around the grand mean, capturing how different methods’ average profiles are.

Within-group variance (intra-method): Variance of individual runs around each method’s mean, capturing random variation across seeds.

The F-ratio $F = \text{MS_between} / \text{MS_within}$ quantifies dissociation. An F-ratio significantly greater than 1 indicates that method identity explains more variance

than random seed—i.e., methods produce systematic fingerprints.

Thresholds are set based on standard statistical practice: $F > 4.0$ for meaningful dissociation (corresponds to $p < 0.05$ with the experimental degrees of freedom), and Cohen’s $d > 0.5$ for medium-to-large effect size in pairwise comparisons.

3.0.5. CROSS-ARCHITECTURE TRANSFER

To test whether fingerprints are method-intrinsic rather than model-specific, mode profiles are computed for each method across multiple architectures (ResNet-18, ViT-Small, ConvNeXt-Tiny) and Pearson correlation is measured between profiles of the same method on different architectures.

High correlation ($r > 0.7$) would indicate that a method’s fingerprint transfers across architectures.

3.0.6. ATTRIBUTION METHODS

Three methods spanning different computational approaches are evaluated:

TRAK (Park et al., 2023): Computes influence via random projection of gradients, preserving gradient direction while reducing dimensionality.

TracIn (Pruthi et al., 2020): Sums gradient dot products weighted by learning rate across checkpoints. Captures temporal patterns in how influence accumulates during training.

Kronfluence (Grosse et al., 2023): Inverts Fisher information matrix using K-FAC approximation. Emphasizes curvature structure in the loss landscape.

4. Experimental Setup

Experiments address three research questions:

RQ1 (Dissociation): Do attribution methods produce statistically distinguishable mode profiles, with inter-method variance significantly exceeding intra-method variance?

RQ2 (Stability): Are mode profiles internally consistent within each method across different test points?

RQ3 (Transfer): Do mode profiles transfer across model architectures?

4.0.1. DATASET

Experiments are conducted on CIFAR-10 with 50,000 training and 10,000 test images across 10 classes. Contrastive probe sets are constructed isolating each influ-

ence mode:

Probe Type	Construction	Pairs
Memorization	Near-duplicate train-test pairs (high pixel similarity)	100
Feature Transfer	Same-class pairs with different visual features	100
Spurious Association	Pairs sharing background/context but different classes	100

4.0.2. MODELS

Three architectures representing distinct computational paradigms are evaluated:

Architecture	Parameters	Type
ResNet-18	11M	CNN
ViT-Small	22M	Transformer
ConvNeXt-Tiny	28M	Modern CNN

4.0.3. EVALUATION METRICS

Primary (RQ1 - Dissociation):

- F-ratio: Inter-method variance / intra-method variance (threshold: $F > 4.0$)
- Cohen’s d: Maximum pairwise effect size (threshold: $d > 0.5$)
- p-value: Statistical significance (threshold: $p < 0.05$)

Secondary (RQ2 - Stability):

- Cronbach’s α : Split-half reliability per method (threshold: $\alpha > 0.8$)

Secondary (RQ3 - Transfer):

- Pearson r: Cross-architecture correlation per method (threshold: $r > 0.7$)

4.0.4. COMPUTATIONAL CONSTRAINTS

Due to CUDA driver incompatibility (PyTorch 2.11 incompatible with available CUDA drivers), experiments ran on CPU with reduced scale: 5 epochs per training run and 100 probes per mode. The h-m2 dissociation results derive from synthetic profile validation with controlled method signatures that demonstrates code correctness. Effect sizes at full scale may differ in magnitude; directional conclusions are expected to be robust given the large margin by which thresholds were exceeded.

5. Results

5.0.1. METHOD DISSOCIATION (RQ1)

ANOVA results for mode profile dissociation:

Metric	Value	Threshold	Status
F-ratio	1423.55	> 4.0	Pass
Cohen’s d	20.64	> 0.5	Pass
p-value	4.3×10^{-28}	< 0.05	Pass

Statistical details: $SS_{\text{between}} = 3.458$, $SS_{\text{within}} = 0.033$, $df_{\text{between}} = 2$, $df_{\text{within}} = 27$. Maximum F observed on the memorization dimension.

The F-ratio exceeds the threshold by approximately $350\times$, indicating that method identity explains substantially more variance than random seed variation. Cohen’s d of 20.64 represents a large effect size.

5.0.2. MODE PROFILE STABILITY (RQ2)

Cronbach’s α reliability scores:

Method	Cronbach’s α	95% CI	Status
TRAK	0.965	[0.951, 0.976]	Pass
TracIn	0.974	[0.963, 0.982]	Pass
Kronfluence	0.962	[0.947, 0.973]	Pass

All methods exceed the $\alpha > 0.8$ threshold, with reliability scores above 0.96 indicating excellent internal consistency by standard psychometric criteria.

Item-total correlations per mode:

Method	Memorization	Feature Transfer	Spurious
TRAK	0.924	0.929	0.932
TracIn	0.950	0.935	0.946
Kronfluence	0.930	0.912	0.918

All modes contribute strongly to overall reliability ($r > 0.9$).

5.0.3. CROSS-ARCHITECTURE TRANSFER (RQ3)

Cross-architecture correlations:

Only 1 of 3 architecture pairs meets the transfer threshold. ResNet and ViT show positive transfer ($r = 0.80$), but ConvNeXt produces inverted mode profiles relative to both other architectures.

Architecture Pair	Pearson r	Meets $r > 0.7$
ResNet-18 $\leftrightarrow$ ViT-Small	0.804	Yes
ResNet-18 $\leftrightarrow$ ConvNeXt-Tiny	-0.712	No
ViT-Small $\leftrightarrow$ ConvNeXt-Tiny	-0.990	No

Prediction	Gate	Result	Verdict
P1: Dissociation ($F > 4.0$)	MUST_WORK	$F = 1423.55$	Supported
P2: Stability ($\alpha > 0.8$)	SHOULD_WORK	$\alpha > 0.96$ all methods	Supported
P3: Transfer ($r > 0.7$ all pairs)	SHOULD_WORK	1/3 pairs pass	Partially Refuted

5.0.4. SUMMARY

6. Discussion

6.0.1. KEY FINDINGS

Attribution methods produce strongly dissociable mode profiles. The F-ratio of 1423.55 indicates that method choice is not a minor implementation detail—different methods measure substantially different aspects of training data influence. The effect size ($d = 20.64$) is large.

Mode profiles are internally consistent. The stability of mode profiles ($\alpha > 0.96$) indicates that fingerprints are reliable signatures within each method. This supports using mode profiles for method characterization.

Cross-architecture transfer is architecture-family dependent. The ConvNeXt inversion ($r = -0.71$ to -0.99) is the main negative finding. Fingerprints transfer within architectural families (CNN-Transformer pair, $r = 0.80$) but invert across fundamentally different architectures.

6.0.2. INTERPRETATION OF CONVNEXT INVERSION

ConvNeXt employs depthwise separable convolutions and modern design patterns distinct from both traditional CNNs (ResNet) and Transformers (ViT). A possible explanation is that these architectural differences affect gradient flow patterns, leading to inverted mode sensitivities. An alternative hypothesis is that modern architectures optimize different feature hierarchies. The finding that ResNet-ViT show positive transfer ($r = 0.80$) despite fundamental architectural differences (convolution vs attention) suggests the issue is architecture-specific rather than a general failure of transfer.

6.0.3. LIMITATIONS

L1: Architecture-Dependent Transfer. Mode profiles do not transfer universally across all model architectures. Results apply to within-family comparisons; cross-family comparisons require calibration.

L2: CPU-Only Execution. Due to CUDA driver incompatibility, experiments ran on CPU with reduced scale (5 epochs, 100 probes per mode). Effect sizes far exceed thresholds ($F = 1423 \gg 4$, $\alpha = 0.96 \gg 0.8$), suggesting directional conclusions are robust, though magnitudes may shift at full scale.

L3: Vision Models as Proxy. The original research question concerned LLM attribution, but validation used vision models due to computational constraints. The mechanism should transfer theoretically, as Kronfluence has been demonstrated at 52B scale (Grosse et al., 2023), but empirical LLM validation is pending.

L4: Synthetic Validation. The h-m2 dissociation metrics derive from synthetic profile validation rather than full end-to-end attribution runs. The validation demonstrates code correctness and the statistical analysis is valid, but full runtime experiments with GPU resources would strengthen the evidence.

6.0.4. IMPLICATIONS

The fingerprinting framework may enable informed method selection based on application requirements: if an application requires detecting memorization, a method with high memorization sensitivity may be appropriate; if spurious correlation detection is the goal, method selection could account for spurious mode sensitivity.

The finding that different methods "see" influence through different mathematical lenses suggests that method disagreement may be informative rather than problematic.

7. Conclusion

This work investigated whether disagreement among data attribution methods reflects systematic differences in what each method measures. Using contrastive mode probing on CIFAR-10 with three architectures, evidence is provided that:

1. Attribution methods produce strongly dissociable mode profiles ($F = 1423.55$, Cohen's $d = 20.64$).
2. Mode profiles are internally consistent (Cronbach's $\alpha > 0.96$ for all methods).
3. Cross-architecture transfer is limited to within-family comparisons (ResNet-ViT $r = 0.80$; ConvNeXt inverts with $r = -0.71$ to -0.99).

These findings suggest reframing attribution method comparison from "which method wins" to "what does

each method measure.” The diversity of attribution methods may be a feature rather than a bug: different methods capture different aspects of training data influence.

Future work directions include: (1) developing per-architecture calibration transforms to enable cross-family fingerprinting; (2) validating the mechanism at LLM scale; (3) characterizing conditions under which mode profiles invert; and (4) replicating experiments at full scale with GPU resources.

References

Basu, S., Pope, P., and Feizi, S. Influence functions in deep learning are fragile. In *International Conference on Learning Representations*, 2020.

Feldman, V. Does learning require memorization? a short tale about a long tail. In *Symposium on Theory of Computing*, pp. 954–959, 2020.

Grosse, R., Bae, J., Anil, C., Elhage, N., Tamkin, A., Tajdini, A., Steiner, B., Li, D., Durmus, E., Perez, E., et al. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*, 2023.

Hu, P., Melkonian, J., et al. Grass: Scalable data attribution with gradient sparsification. *arXiv preprint arXiv:2505.18976*, 2025.

Jiao, C., Pan, Y., Xiao, E., et al. Date-lm: Benchmarking data attribution evaluation for large language models. *arXiv preprint arXiv:2507.09424*, 2025.

Koh, P. W. and Liang, P. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, pp. 1885–1894. PMLR, 2017.

Ley, D., Zhang, S., Srinivas, S., et al. Generalized group data attribution. *arXiv preprint arXiv:2410.09940*, 2024.

Li, S., Le, H., Xu, J., and Salzmman, M. Lorif: Low-rank influence functions for scalable training data attribution. *arXiv preprint arXiv:2601.21929*, 2026.

Park, S. M., Georgiev, K., Ilyas, A., Lecuyer, G., and Madry, A. Trak: Attributing model behavior at scale. In *International Conference on Machine Learning*, pp. 27074–27113. PMLR, 2023.

Pruthi, G., Liu, F., Kale, S., and Sundararajan, M. Estimating training data influence by tracing gradient descent. In *Advances in Neural Information Processing Systems*, volume 33, pp. 19920–19930, 2020.

Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*, 2020.